FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
CONCEPT RECAPITULATION TEST – II
PAPER –1
TEST DATE: 01-05-2024

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 –03, 18 – 20, 35 – 37):* This section contains **NINE (9)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A *(04 – 07, 21 – 24, 38 – 41)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A *(08 – 11, 25 – 28, 42 – 45):* This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (P), (Q), (R) and (S) and **FIVE** statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B *(12 – 17, 29 – 34, 46 – 51):* This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

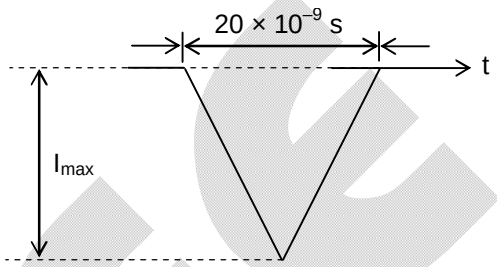
Physics

PART – I

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

1. A photomultiplier tube is to be used to detect light pulses each of which consists of a small but fixed number of photons. The average photoelectric efficiency is 10%. That is photon has 10% probability; of causing the emission of a detectable photoelectron. Assume the photomultiplier gain is 10^6 and that the output current as a function of time can be approximated as shown in figure.



I_{max} when averaged over many pulses is 80 μA. Then which of the following statements is/are true.

(A) The charge carried by on pulse is 8×10^{-13} C.

(B) Number of photoelectrons emitted per light pulse is 5.

(C) Number of photons in one light pulse is 50.

(D) Number of electrons carried by one pulse is 5×10^5.

2. Consider a ray of light going from A to B. Let the ray traverse, in going from A to B, distances $S_1, S_2, S_3, \ldots\ldots\ldots\ldots S_m$ in media of indices $n_1, n_2, n_3, \ldots\ldots\ldots\ldots n_m$ respectively.

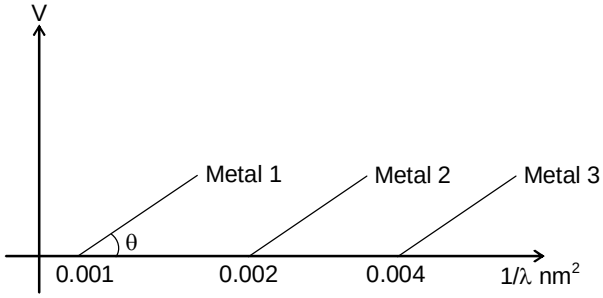
(A) Total time take by light $= \frac{1}{c}\sum_{i=1}^{m} n_i s_i$

(B) Total time taken by light $= \frac{1}{C}\sum_{i=1}^{m} s_i$

(C) Optical path length is (O.P.I.) = $\sum_{i=1} n_i s_i$

(D) For inhomogeneous media the $\text{O.P.I.} = \int_A^B n(s)\, ds$ and the ray travels along 'stationary pathways'.

3. The graph between $\frac{1}{\lambda}$ and stopping potential (V) of three metals having work function ϕ_1, ϕ_2 and ϕ_3 in an experiment of photoelectric effect is plotted as shown in the figure. Which of the following statement(s) is/are correct? [Here λ is the wavelength of the incident ray]

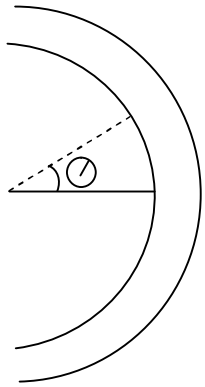


(A) Ratio of work functions $\phi_1 : \phi_2 : \phi_3 = 1 : 2 : 4$.
(B) Ratio of work functions $\phi_1 : \phi_2 : \phi_3 = 4 : 2 : 1$.
(C) $\tan\theta$ is directly proportional to $\frac{hc}{e}$, where h is Planck's constant and C is the speed of light.
(D) None of the above.

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

4. A modern grand-prix racing car of mass m is travelling on a flat track in a circular arc of radius R with a speed v. If the coefficient of static friction between the tyres and the track is μ_s, then the magnitude of negative lift F_L acting downwards on the car is (Assume forces on the four tyres are identical and g = acceleration due to gravity)

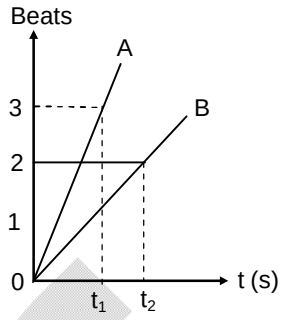


(A) $m\left[\frac{v^2}{\mu_s R} - g\right]$
(B) $-m\left[g + \frac{v^2}{\mu_s R}\right]$
(C) $m\left[\frac{v^2}{\mu_s R} + g\right]$
(D) $m\left[g - \frac{v^2}{\mu_s R}\right]$

5. A beaker contains a fluid of density ρ kg/m^3, specific heat S J/kg°C and viscosity η. The beaker is filled up to height h. To estimate the rate of heat transfer per unit area (Q/A) by convection when beaker is put on the hot plate, a student proposes that it should depend on η. $\left(\frac{S\Delta\theta}{h}\right)$ and $\left(\frac{1}{\rho g}\right)$ when $\Delta\theta$ (in °C) is the difference in the temperature between the bottom and top of the fluid. In that situation, the correct option for (Q/A) is

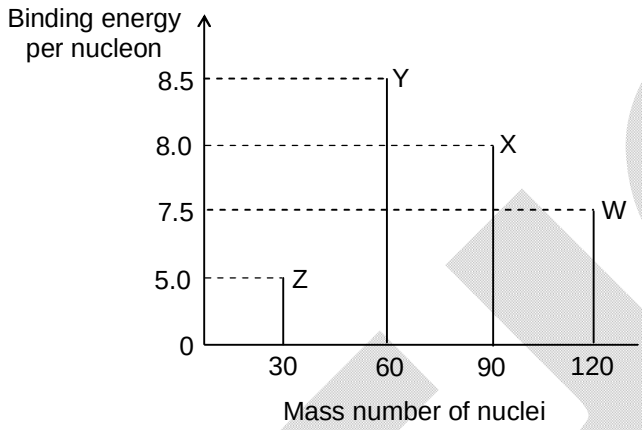
(A) $\eta\frac{S\Delta\theta}{h}$
(B) $\eta\left(\frac{S\Delta\theta}{h}\right)\left[\frac{1}{\rho g}\right]$
(C) $\frac{S\Delta\theta}{\eta h}$
(D) $\left(\frac{S\Delta\theta}{\eta h}\right)\left[\frac{1}{\rho g}\right]$

6. Two tuning forks P and Q are vibrated together. The number of beats produced are represented by the straight line OA in the graph. After loading Q with wax again these are vibrated together and the beats produced are represented by the line OB. If the frequency of P is 341 Hz, the frequency of Q will be



(A) 341 Hz (B) 338 Hz
(C) 344 Hz (D) 518 Hz

7. Binding energy per nucleon vs mass number curve for nuclei is shown in the figure. W, X, Y and Z are four nuclei indicated on the curve. The process that would release energy is

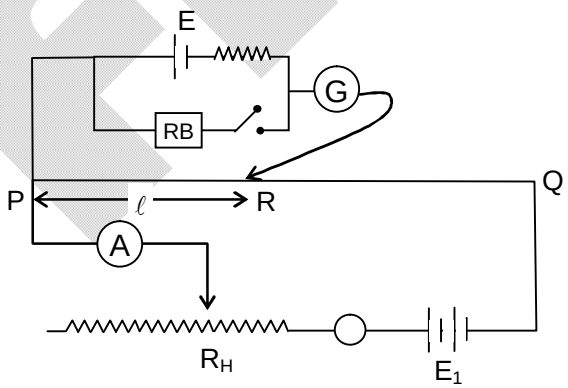


(A) $Y \rightarrow 2Z$ (B) $W \rightarrow X + Z$
(C) $W \rightarrow 2Y$ (D) $X \rightarrow Y + Z$

SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

8. A potentiometer experiment setup is shown in the figure. R is the point where there is no deflection in the galvanometer. E_1 is the source. Match the following:



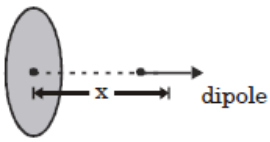
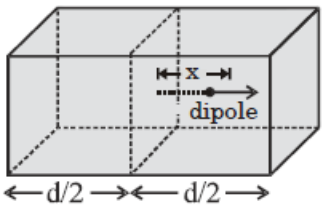
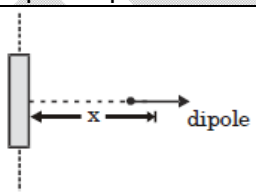
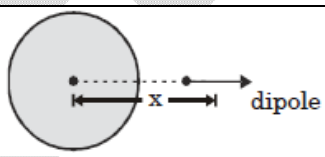
List-I		List-II	
(P)	If only emf of battery E_1 is increased.	(1)	Point R will shift to left.
(Q)	If only resistance offered by rheostat is increased.	(2)	Point R will shift to right.
(R)	If resistance box (RB) is connected and resistance of RB is increased.	(3)	Point R will remain unchanged.
(S)	An identical battery is connected in parallel to E	(4)	Point R may shift to left or right.
		(5)	Cannot say

The correct option is:

(A) P – 3 ; Q – 4 ; R – 2 ; S – 1 (B) P – 1 ; Q – 2 ; R – 1 ; S – 3

(C) P – 4 ; Q – 3 ; R – 1 ; S – 2 (D) P – 2 ; Q – 1 ; R – 4 ; S – 3

9. List-I shows different charge distributions and short electric dipole at a distance x from the charge distributions. List-II gives the dependence of force acting on the dipole as of function of x.

List-I		List-II	
(P)	 Uniformly charged thin infinite disc.	(1)	x^{-2}
(Q)	 Uniformly charged thick infinite sheet. Dipole is placed inside the sheet.	(2)	x^{-3}
(R)	 Uniform infinite line of charge.	(3)	x^{0}
(S)	 Uniformly charged sphere. Dipole is placed outside the sphere.	(4)	$x^{-\infty}$
		(5)	

The correct option is:

(A) P $\rightarrow$ 4 ; Q $\rightarrow$ 1 ; R $\rightarrow$ 3 ; S $\rightarrow$ 2 (B) P $\rightarrow$ 4 ; Q $\rightarrow$ 3 ; R $\rightarrow$ 1 ; S $\rightarrow$ 2

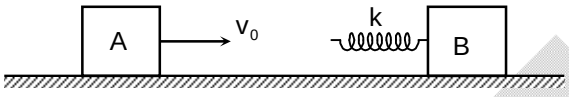
(C) P $\rightarrow$ 1 ; Q $\rightarrow$ 2 ; R $\rightarrow$ 4 ; S $\rightarrow$ 3 (D) P $\rightarrow$ 3 ; Q $\rightarrow$ 4 ; R $\rightarrow$ 1 ; S $\rightarrow$ 2

10. A particle of mass 'm' and charge 'q' is projected from origin, with velocity $\vec{V} = V_0\hat{i}$ and magnetic field. $\vec{B} = -(B_0x)\hat{k}$. Match the correct option if $\frac{m}{q_0} = 1$ kg/c, V_0 = 2 m/s and B_0 = 1T.

List-I		List-II	
(P)	Radius of curvature at point of projection	(1)	2
(Q)	Radius of curvature when charge is at maximum x-coordinate	(2)	1
(R)	Maximum x-coordinate	(3)	∞
(S)	Maximum radius of curvature of path followed.	(4)	1/2
		(5)	0

(A) P – 2 ; Q – 2 ; R – 2 ; S – 3
(B) P – 2 ; Q – 1 ; R – 2 ; S – 3
(C) P – 3 ; Q – 2 ; R – 1 ; S – 2
(D) P – 3 ; Q – 2 ; R – 1 ; S – 3

11. Consider two blocks as shown in the figure. The block A (mass m) is moving with velocity v_0 and collides elastically with the ideal spring having spring constant k attached to block B (mass 2 m). There is no friction. Match List-I with List-II in ground frame.



List - I		List - II	
(P)	Maximum compression of the spring	(1)	$\sqrt{2mv_0^2/3k}$
(Q)	Relative velocity of blocks A and B at the time when spring is maximum compressed	(2)	$mv_0^2/3$
(R)	Velocity of A	(3)	Zero
(S)	Maximum energy stored in the spring	(4)	$v_0/3$
		(5)	v_0

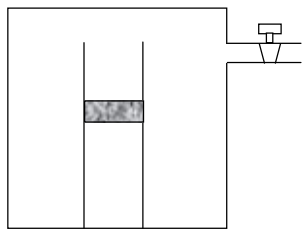
(A) P – 1 ; Q – 3 ; R – 4 ; S – 5
(B) P – 2 ; Q – 1 ; R – 5 ; S – 4
(C) P – 1 ; Q – 3 ; R – 4 ; S – 2
(D) P – 3 ; Q – 4 ; R – 3 ; S – 3

SECTION – B

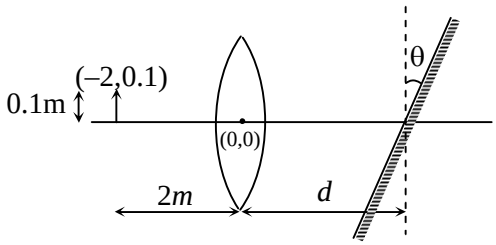
(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

12. An ideal gas is kept in a long cylindrical vessel fitted with a frictionless piton of cross-sectional area 10 cm^2 and weight 1 kg. The vessel itself is kept in a big chamber containing air at atmospheric pressure 100 kPa.
The length of the gas column is 20 cm. If the chamber is now completely evacuated by an exhaust pump, 110x cm will be the length of the gas column. Assume the temperature remain constant through the process. Find the value of 'x'.

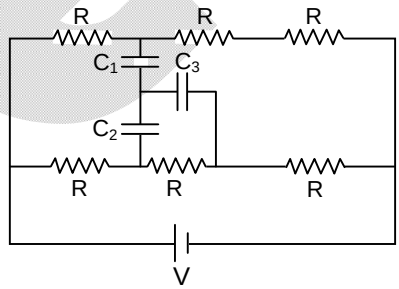


13. A convex lens of focal length 1.5 m is placed in a system of coordinate axis such that its optical centre is at origin and principal axis coinciding with the x-axis. An object and a plane mirror are arranged on the principal axis as shown in figure. Find the value of d (in m) so that y-coordinate of final image (after refraction and reflection) is 0.3 m. (Take tan $\theta = 0.3$)

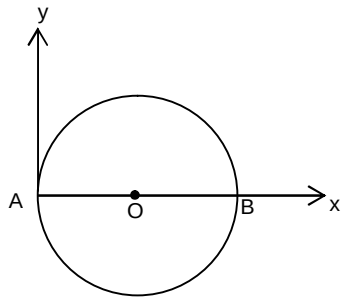


14. A solid billiard ball of radius 'R' and mass 'm' initially at rest is given a sharp impulse by a cue' held horizontally at a distance $\frac{R}{2}$ above the centre. Just after the impulse, the velocity of centre of mass of the ball is V = 10 m/s. The coefficient of friction between ball and table is $\mu = \frac{1}{2}$. The ball starts rolling without slipping t seconds after impulse is given. Find value of $\frac{1}{t}$ (in sec).

15. In the shown circuit, all three capacitor are identical and have capacitance CμF each. Each resistor has resistance of R Ω. An ideal cell of emf V volts in connected as shown.
If the magnitude of potential difference across capacitor C_3 in steady state is $\frac{a}{b}V$ then value of a × b is:



16. A man is running with constant speed of $v = 10\sqrt{5}$ m/s on a horizontal track of radius R = 20 m as shown in figure. At position A man launches a stone in space (without changing his own speed) so that he can catch stone at B (diametrically opposite to A). The speed of launch will be approximately 4K m/sec. Then find the value of K.
(take $\pi^2 = 10$ and g = 10 m/s^2)



17. A closed organ pipe of length ℓ is sounded together with another closed organ pipe of length
ℓ + x (x << ℓ) both in fundamental mode.
If V = speed of sound, the beat frequency heard is f. Find $10f_0$
(If V = 320 m/s, x = 2 cm, ℓ = 2 m)

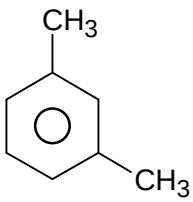
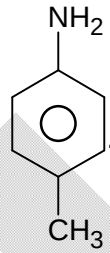
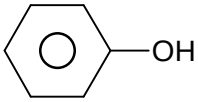
Chemistry

PART – II

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

18. Which of the following statement(s) is/are correct when a mixture of $CaCl_2$ and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4?
(A) Deep red vapours are evolved
(B) CrO_2Cl_2 is formed
(C) The vapours so evolved turn NaOH solution yellow
(D) CaF_2 can also be used in place of $CaCl_2$ in given mixture to observe similar colour changes

19. A mixture contains  and  . Which of the following substance(s) is/are required together to differentiate the components of the mixture?
(A) $NaNO_2$
(B) HCl
(C) 
(D) $NaHCO_3$

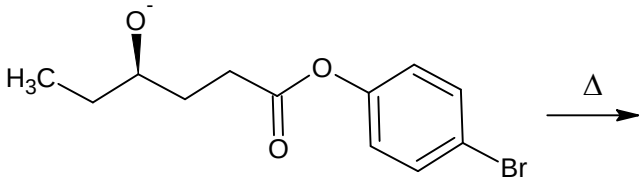
20. Which of the following statements is/are true about an azeotropic mixture?
(A) An azeotropic mixture boils at constant temperature
(B) The composition of an azeotropic mixture changes on distillation
(C) An azeotropic solution of two liquids has a boiling point lower than that of pure components for a non-ideal solution forming minimum boiling point azeotrope.
(D) An azeotropic solution of two liquids has a boiling point higher than that of pure components for a non-ideal solution forming maximum boiling point azeotrope.

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

21. If a hydrogen molecule absorbs a photon having energy exactly equal to the energy difference in between antibonding σ^*_{1S} MO and bonding σ_{1S} MO of H_2 molecule, what happens?
(A) The H_2 molecule will ionize
(B) The H_2 molecule will get stabilized
(C) The bond between molecule breaks and dissociated atoms have some kinetic energy
(D) The bond between molecule breaks and dissociated atoms have zero kinetic energy

22. Which of the following compound of lead undergoes redox reaction on heating?
(A) $PbCO_3$ (B) PbO
(C) PbI_4 (D) $PbSO_4$

23. 

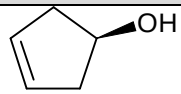
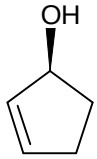
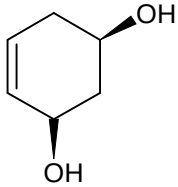
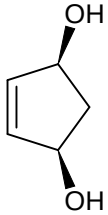
Which of the statement is **NOT CORRECT** regarding the product formed in the above reaction?
(A) If Br is replaced with NO_2, the rate of reaction increases
(B) Inversion of configuration takes place.
(C) Net reaction is nucleophilic addition elimination.
(D) Configuration at chiral carbon does not change

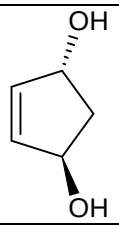
24. Which of the following molecule have shortest Hg – Hg bond length?
(A) Hg_2F_2 (B) Hg_2Cl_2
(C) Hg_2Br_2 (D) Hg_2I_2

SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

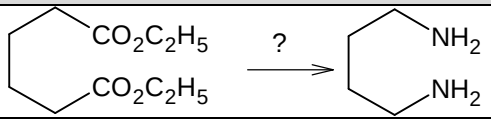
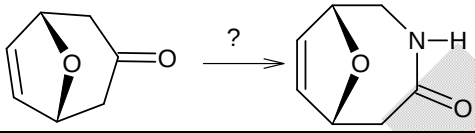
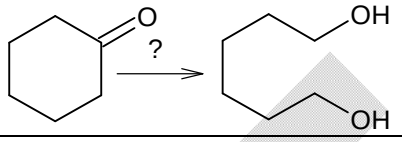
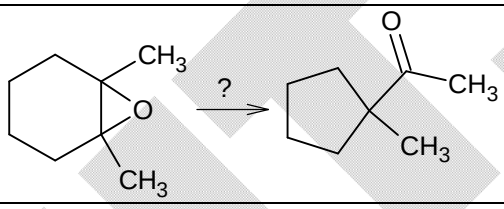
25. In list – I, statements are given and in list- II, structures are provided. Match the statement given in Column I with corresponding structures shown in list – II

List – I		List – II	
(P)	An optically active compound that is oxidized by MnO_2 to an optically inactive compound.	(1)	
(Q)	An optically active compound that is oxidized by MnO_2 to an optically active compound	(2)	
(R)	An optically inactive compound that is oxidized by MnO_2 to an optically inactive compound	(3)	
(S)	A compound that is not oxidized by MnO_2	(4)	

		(5)	

(A) P → 5, Q → 3, R → 4, S → 1 (B) P → 5, Q → 2, R → 3, S → 4
(C) P → 4, Q → 5, R → 2, S → 1 (D) P → 2, Q → 1, R → 4, S → 3

26. Synthetic transformation A to D are shown here. Each of these has been carried out by a sequence of reactions, one of which involves a molecular rearrangement. A list of five possible procedures is given to the right of equations. Your job is to pick the sequence (or single reaction) that best achieves the selected synthesis.

List – I		List – II	
(P)		(1)	$(i)C_6H_5CO_3H$ in CH_2Cl_2; $(ii)LiAlH_4$ in ether; (iii) mild acidification
(Q)		(2)	$(i)LiAlH_4$ in ether; $(ii)H_2O$; $(iii)C_6H_5CO_3H$ in CH_2Cl_2
(R)		(3)	H_2SO_4 (catalyst) in CH_2Cl_2 solvent
(S)		(4)	(i) NH_2OH (oxime formation) (ii) strong acid or PCl_5 (Beckmann rearrangement)
		(5)	N_2H_4 (excess) (ii) HNO_2; $H_3O^{(+)}$; 5°C

(A) P → 1, Q → 3, R → 5, S → 2 (B) P → 1, Q → 4, R → 3, S → 2
(C) P → 5, Q → 4, R → 1, S → 3 (D) P → 4, Q → 3, R → 2, S → 1

27. Match the items given in list – I with that in list – II

List – I		List – II	
(P)	Raoult's law	(1)	Effect of pressure on the solubility of gas in liquid
(Q)	Henry's law	(2)	$\frac{P^o - P_s}{P^o} = \chi_{\text{non volatile solute}}$
(R)	Ethyl alcohol + water	(3)	Ideal solution
(S)	Benzene + Toluene	(4)	Azeotropic mixture
		(5)	Non-ideal solution

(A) P → 1, Q → 3, R → 5, S → 2 (B) P → 2, Q → 1, R → 4, S → 3
(C) P → 4, Q → 5, R → 2, S → 1 (D) P → 2, Q → 4, R → 5, S → 1

28. Match the elements mentioned in list-I with their properties mentioned in list-II.

List – I		List – II	
(P)	Zn(Z = 30)	(1)	Forms colourless compound in +3 oxidation state
(Q)	Fe(Z = 26)	(2)	The spin-value is zero in +2 oxidation state
(R)	Sc(Z = 21)	(3)	Magnetic moment (spin - only) is $2\sqrt{6}$ B.M in +2 oxidation state
(S)	Ag (Z = 47)	(4)	Diamagnetic in +1 oxidation state
		(5)	Paramagnetic in +1 oxidation state

(A) P $\rightarrow$ 2, Q $\rightarrow$ 3, R $\rightarrow$ 1, S $\rightarrow$ 4 (B) P $\rightarrow$ 2, Q $\rightarrow$ 1, R $\rightarrow$ 4, S $\rightarrow$ 3
(C) P $\rightarrow$ 4, Q $\rightarrow$ 5, R $\rightarrow$ 2, S $\rightarrow$ 1 (D) P $\rightarrow$ 5, Q $\rightarrow$ 4, R $\rightarrow$ 2, S $\rightarrow$ 1

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

29. $ZnFe_2O_4$ adopt spinel structure. In the structure, the divalent cation occupied $\frac{1}{x}$ part of available tetrahedral voids and trivalent cation occupied $\frac{1}{y}$ part of available octahedral voids. The ratio of $\frac{1}{y}:\frac{1}{x}$ is_____.

30. 4.48 litre of an ideal gas at STP requires 12 calorie to raise the temperature by 15°C at constant volume. What is the value of C_P of the gas in calorie unit?

31. $\underset{\text{metal chloride}}{(P)} \xrightarrow{NaOH} \underset{\text{green ppt.}}{(Q)\downarrow} \xrightarrow{\text{Excess NaOH}} \underset{\text{green solution}}{(R)}$

$\underset{\text{Solution}}{(R)} \xrightarrow{H_2O_2} \underset{\text{yellow solution}}{(S)} \xrightarrow[BaCl_2]{CH_3COOH} \underset{\text{yellow ppt.}}{(T)\downarrow}$

If the molar mass of P is x g/mol, then value of 2x is

32. A metal oxide Z_2O_3 can be reduced by hydrogen to give free metal and water. 0.1596 gm metal oxide requires 6 mg hydrogen. If the atomic weight of metal is x, then value of 10x is

33. The dipole moment of a polar covalent molecule XY is 3 Debye. The bond length of XY is 1.25×10^{-8} cm. If the percentage ionic character in the molecule is expressed as X%, then the value of 'X' will be [Given: Charge of electron = -4.8×10^{-10} esu]

34. Acetic acid concentration in commercial vinegar was analyzed by titrating it against 1.5 M NaOH solution. A 20 mL vinegar sample required 40 mL of titrant to give end point. The concentration of acetic acid in the vinegar in mol L^{-1} is

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

35. Solution of the differential equation $\left(1-x^2\right)\frac{dy}{dx}+xy=ax$ where $a\in\mathbb{R}$, is

(A) a conic which is an ellipse or a hyperbola with principal axes parallel to coordinate axes
(B) centre of the conic is (0, a)
(C) length of one of the principal axes is 1
(D) length of one of the principal axes is equal to 2

36. Let z_1, z_2, z_3 be non – zero complex numbers satisfying the equation $z^4 = iz$. Which of the following statement(s) is/are correct?

(A) The complex number having least positive principal argument is $\frac{\sqrt{3}}{2}+\frac{i}{2}$

(B) $\sum_{k=1}^{3}\text{Arg}\left(z_k\right)=\frac{\pi}{2}$

(C) Centroid of the triangle formed by z_1, z_2 and z_3 is $\left(\frac{1}{\sqrt{3}},\frac{-1}{3}\right)$

(D) Area of triangle formed by z_1, z_2 and z_3 is $\frac{3\sqrt{3}}{2}$.

37. D is a 3×3 diagonal matrix. Which of the following statement(s) is/are NOT true?
(A) $D' = D$
(B) $AD = DA$ for every matrix A of order 3×3
(C) D^{-1} if exists is a scalar matrix
(D) none of the above

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

38. m, n are integers with $0<n<m$. A is the point (m, n) on the cartesian plane. B is the reflection of A in the line $y=x$. C is the reflection of B in the y – axis, D is the reflection of C in the x – axis and E is the reflection of D in the y – axis. The area of the pentagon ABCDE is

(A) $2m(m+n)$ (B) $m(m+3n)$
(C) $m(2m+3n)$ (D) $2m(m+3n)$

39. If $a<0$ and $a-b+c=a-3b+9c=0$, then the quadratic expression $y=ax^2+bx+c$ will attain its maximum value at $x=k$, then k is

(A) $-\frac{1}{3}$ (B) $-\frac{2}{3}$

(C) –1 (D) $\frac{-4}{3}$

40. If $\vec{a},\vec{b}$ and $\vec{c}$ be three vectors having magnitudes 1, 1 and 2 respectively. If $\vec{a}\times\left(\vec{a}\times\vec{c}\right)+\vec{b}=0$, then the acute angle between $\vec{a}$ and $\vec{c}$ is:

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{3}$ (D) $\frac{5\pi}{12}$

41. For each positive integer n, consider the point P in the first quadrant with abscissa n on the curve $y^2-x^2=1$. If d_n represents the shortest distance from the point P to the line $y=x$ then $\lim\limits_{n\to\infty}\left(n.d_n\right)$ has the value equal to

(A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{2}}$ (D) 0

SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

42.

List - I		List - II	
(P)	In an A.P. series containing 99 terms, the sum of all the odd numbered terms is 2550. The sum of all the 99 terms of the A.P. is	(1)	5010
(Q)	f is a function for which $f(1)=1$ and $f(n)=n+f(n-1)$ for each $n\geq 2$, $n\in\mathbb{N}$. The value of $f(100)$ is	(2)	5049
(R)	Suppose, $f(n)=\log_2(3).\log_3(4).\log_4(5).....\log_{n-1}(n)$ then the sum $\sum_{k=2}^{100} f\left(2^k\right)$ equals	(3)	5050
(S)	Concentric circles of radii 1, 2, 3,….100 cm are drawn. The interior of the smallest circle is coloured red and the annular regions are coloured alternatively green and red so that no two adjacent regions are of the same color. The total area of the green regions in sq. cm is $k\pi$ then 'k' equals	(4)	5100
		(5)	5051

(A) P $\rightarrow$ 2, Q $\rightarrow$3, R $\rightarrow$2, S $\rightarrow$3 (B) P $\rightarrow$ 3, Q $\rightarrow$2, R $\rightarrow$4, S $\rightarrow$1
(C) P $\rightarrow$ 2, Q $\rightarrow$5, R $\rightarrow$1, S $\rightarrow$4 (D) P $\rightarrow$ 3, Q $\rightarrow$5, R $\rightarrow$4, S $\rightarrow$2

43.

List - I		List - II	
(P)	Let ω be a non – real cube root of unity then the number of distinct elements in the set $\left\{\left(1+\omega+\omega^2+.....\omega^n\right)^m \mid m,n\in\mathbb{N}\right\}$ is	(1)	4
(Q)	Let $1,\omega,\omega^2$ be the cube roots of unity. The least possible degree of a polynomial with real coefficients having roots $2\omega, 2+3\omega, 2+3\omega^2, 2-\omega-\omega^2$, is	(2)	5
(R)	Number of complex numbers z satisfying the equation $\bar{z}=iz^2$, where $i=\sqrt{-1}$ is equal to	(3)	6
(S)	$\alpha=6+4i, \beta=2+4i$ are two complex numbers. Then complex number z satisfying $\arg\left(\frac{z-\alpha}{z-\beta}\right)=\frac{\pi}{6}$ moves on the major segment of a circle of radius equal to	(4)	7
		(5)	8

(A) P $\rightarrow$ 2, Q $\rightarrow$3, R $\rightarrow$4, S $\rightarrow$5 (B) P $\rightarrow$ 4, Q $\rightarrow$2, R $\rightarrow$1, S $\rightarrow$1
(C) P $\rightarrow$ 2, Q $\rightarrow$5, R $\rightarrow$1, S $\rightarrow$4 (D) P $\rightarrow$ 3, Q $\rightarrow$5, R $\rightarrow$4, S $\rightarrow$2

44.

List - I		List - II	
(P)	Order of differential equation whose general solution is given by $y=(c_1+c_2)\cos(x+c_3)-c_4e^{x+c_5}+C_5\sin x$ where c_1,c_2,c_3,c_4,c_5 are arbitrary constants, is	(1)	1
(Q)	Order of differential equation formed by eliminating the constants from $y=a\sin^2x+b\cos^2x+c\sin 2x+d\cos 2x+e\sin x$, where a, b, c, d, e are arbitrary constants, is	(2)	2
(R)	The degree of the equation $\frac{d^2y}{dx^2}+\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}=0$ is	(3)	3
(S)	Order of the differential equation whose solution is $y=cx+c^2-3c^{3/2}+2$, where C is a parameter is	(4)	4
		(5)	5

(A) P $\rightarrow$ 2, Q $\rightarrow$3, R $\rightarrow$4, S $\rightarrow$5 (B) P $\rightarrow$ 3, Q $\rightarrow$4, R $\rightarrow$2, S $\rightarrow$1
(C) P $\rightarrow$ 2, Q $\rightarrow$5, R $\rightarrow$1, S $\rightarrow$4 (D) P $\rightarrow$ 3, Q $\rightarrow$5, R $\rightarrow$4, S $\rightarrow$2

45.

List - I		List - II	
(P)	The number of points where the function f is continuous but not differentiable is, $f(x)=\begin{cases}1+\left[\cos\frac{\pi x}{2}\right], & 1<x\le 2\\ 1-\{x\}, & 0\le x<1\\ \lvert\sin\pi x\rvert, & -1\le x<0\\ 0, & x=1\end{cases}$ ($\{x\}$ = fractional part of x and $[x]$ = integral part of x)	(1)	0
(Q)	If $f(x)=\begin{cases}x^2e^{\frac{1}{x}}, & x\neq 0\\ 0, & x=0\end{cases}$, then $f'(0^-)=$	(2)	1
(R)	The number of points at which $g(x)=\frac{1}{1+\frac{2}{f(x)}}$ is not differentiable where $f(x)=\frac{1}{1+\frac{1}{x}}$, is	(3)	2
(S)	The number of points where tangent does not exist for the curve $y=\text{sgn}(x^2-1)$ is	(4)	3
		(5)	4

(A) P $\to$ 2, Q $\to$1, R $\to$4, S $\to$3 (B) P $\to$ 3, Q $\to$2, R $\to$4, S $\to$1
(C) P $\to$ 2, Q $\to$5, R $\to$1, S $\to$4 (D) P $\to$ 3, Q $\to$5, R $\to$4, S $\to$2

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

46. Let $a=\left(4^{\frac{1}{401}}-1\right)$ and for each $n\ge 2$, let $b_n={}^nC_1+{}^nC_2\,a+{}^nC_3.a^2+{}^nC_4.a^3+...+{}^nC_n.a^{n-1}$.
If the value of $(b_{2006}-b_{2005})$ is 4^k then the value of k is

47. If x_i, $i=1,2,3,....,m$ are non – negative integers satisfying the inequality $x_1+x_2+x_3+....+x_m\le 20$ and $N(m)$ is the number of solutions of the inequality then the value of $\frac{N(5)}{N(4)}$ is _____

48. T_1, T_2 and T_3 are three different taxies, each having three duly numbered seats for the passengers. Seven passengers are to be accommodated in these taxies. If probability that taxies T_1 and T_2 are used to their full capacity is $\frac{m}{n}$, $m,n\in\mathbb{N}$, then the least positive value of $n-4m$ is

49. Let $u=\int_0^1 \frac{\log(x+1)}{x^2+1}dx$ and $v=\int_0^{\pi/2}\log(\sin 2x)dx$ then the value of $\frac{-v}{u}$ is _____

50. Considering only the principal values of the inverse trigonometric functions, the value of $\frac{3}{2}\cos^{-1}\sqrt{\frac{2}{2+\pi^2}}+\frac{1}{4}\sin^{-1}\left(\frac{2\sqrt{2}\pi}{2+\pi^2}\right)+\tan^{-1}\left(\frac{\sqrt{2}}{\pi}\right)$ is $\frac{k\pi}{4}$ then k is_____ .

51. Rectangle ABCD has area 200. An ellipse with area 200π passes through A and C and has foci at B and D. Then the perimeter of the rectangle is _____

FIITJEE